Practice by HMHDental, book followed is KLM Human Embryo and Langman's

Embryology Professional Examination: *Weeks 1–3*

1. Human chorionic gonadotropin (hCG) produced by the syncytiotrophoblast can be detected in maternal blood by radioimmunoassay as early as which day of development?

a. Day 2
b. Day 6
c. Day 12
d. Day 14
e. Day 8

2. In a clinical setting, a highly sensitive pregnancy test using a maternal urine sample can reliably show a positive result on which day?

a. Day 2
b. Day 6
c. Day 8
d. Day 10
e. Day 14

3. The flat, bilaminar embryonic disc formed during the second week of development consists of which two specific cell layers?

a. Syncytiotrophoblast and Cytotrophoblast
b. Ectoderm and Endoderm
c. Somatopleure and Splanchnopleure
d. Epiblast and Hypoblast
e. Amnion and Yolk sac

4. The definitive process of gastrulation, which establishes all three primary germ layers in the embryo, is initiated by the appearance of the:

a. Neural tube
b. Notochord
c. Prechordal plate
d. Prochordal plate
e. Primitive streak

5. A newborn is diagnosed with a large sacrococcygeal teratoma containing hair, teeth, and muscle tissue. This tumor arises from the persistence of pluripotent remnants of the:

a. Primitive streak
b. Notochord
c. Neural crest cells
d. Prechordal plate
e. Oropharyngeal membrane

6. If a blastocyst fails to implant in the body of the uterus and instead attaches within the lower uterine segment near the internal os, the condition is termed:

a. Ectopic tubal pregnancy
b. Abdominal pregnancy
c. Placenta previa
d. Hydatidiform mole
e. Sirenomelia

7. Maternal blood first enters the primitive uteroplacental circulation by flowing into isolated trophoblastic lacunae during which specific timeframe?
a. Day 2
b. Day 6
c. Day 9
d. Day 15
e. Day 21
8. Which architectural component of the blastocyst is directly responsible for invading the endometrial stroma and secreting proteolytic enzymes?
a. Epiblast
b. Hypoblast
c. Cytotrophoblast
d. Embryoblast
e. Syncytiotrophoblast
9. The primary organizer for the head region, which induces the overlying ectoderm to form forebrain structures, is the:
a. Primitive node
b. Prechordal plate
c. Cloacal membrane
d. Oropharyngeal membrane
e. Notochord
10. By the end of the first week of human development, the trophoblast differentiates into two distinct layers, which are the:
a. Epiblast and Hypoblast
b. Cytotrophoblast and Syncytiotrophoblast
c. Somatopleure and Splanchnopleure
d. Ectoderm and Endoderm
e. Amnion and Chorion
11. The extraembryonic coelom, which forms a large fluid-filled space surrounding the amniotic sac and primitive yolk sac, is also called the:
a. Amniotic cavity
b. Chorionic cavity
c. Blastocyst cavity
d. Intraembryonic coelom
e. Exocoelomic cavity
12. The first morphological evidence of the future nervous system appears in the third week as a thickened area of ectoderm known as the:
a. Neural plate
b. Neural tube
c. Neural crest
d. Notochord
e. Somite
13. The transition from a solid morula into a hollow blastocyst occurs when fluid enters the intercellular spaces, creating a single large cavity called the:
a. Amniotic cavity
b. Yolk sac
c. Blastocyst cavity
d. Chorionic cavity

e. Exocoelomic cavity

14. A solid, spherical ball of blastomeres formed by early cleavage divisions prior to blastocyst formation typically consists of how many cells?

a. 2 to 4 cells
b. 12 to 16 cells
c. 32 to 64 cells
d. 100 to 200 cells
e. Exactly 8 cells

15. Implantation of the human blastocyst into the epithelial lining of the endometrium normally begins on which day following fertilization?

a. Day 2
b. Day 4
c. Day 6
d. Day 10
e. Day 14

16. The cellular reaction where endometrial stromal cells become polyhedral, hyperplastic, and loaded with glycogen and lipids at the site of implantation is the:

a. Acrosome reaction
b. Cortical reaction
c. Zona reaction
d. Capacitation
e. Decidual reaction

17. A chorionic villus that consists structurally of a central core of extraembryonic mesoderm covered completely by a layer of cytotrophoblast and syncytiotrophoblast is classified as a:

a. Primary villus
b. Secondary villus
c. Tertiary villus
d. Stem villus
e. Anchoring villus

18. The process of gastrulation involves the invagination of cells from which specific layer through the primitive streak?

a. Epiblast
b. Hypoblast
c. Cytotrophoblast
d. Syncytiotrophoblast
e. Extraembryonic mesoderm

19. Severe caudal dysgenesis resulting from inadequate mesoderm formation in the caudal-most region of the embryo is clinically known as:

a. Sirenomelia
b. Anencephaly
c. Spina bifida
d. Gastroschisis
e. Omphalocele

20. The amniotic cavity begins its structural development as a small, isolated space appearing within the:

a. Hypoblast
b. Cytotrophoblast

c. Syncytiotrophoblast
d. Epiblast
e. Extraembryonic mesoderm

21. For successful attachment and implantation to take place, the expanding blastocyst must first shed which surrounding acellular protective layer?

a. Zona pellucida
b. Corona radiata
c. Plasma membrane
d. Cytotrophoblast
e. Vitelline membrane

22. Which specific germ layer gives rise to the epithelial lining of the gastrointestinal tract, urinary bladder, and respiratory system?

a. Surface ectoderm
b. Neuroectoderm
c. Mesoderm
d. Endoderm
e. Neural crest

23. The connecting stalk, which suspends the developing embryo inside the extraembryonic coelom, serves as the precursor to the:

a. Placenta
b. Amniotic sac
c. Umbilical cord
d. Allantois
e. Chorion frondosum

24. The fundamental structural axis of the embryo, which serves as the basis for the axial skeleton and induces neurulation, is the:

a. Primitive streak
b. Primitive node
c. Prechordal plate
d. Notochord
e. Prechordal mesoderm

25. A primary chorionic villus is histologically defined by the presence of:

a. Syncytiotrophoblast cells only
b. A mesodermal core with capillary loops
c. Fetal blood vessels only
d. A central core of cytotrophoblast covered by syncytiotrophoblast
e. Maternal blood sinusoids only

26. The exocoelomic membrane (Heuser's membrane) combines with the hypoblast to form the lining of the:

a. Amniotic cavity
b. Primary yolk sac
c. Chorionic cavity
d. Extraembryonic coelom
e. Intraembryonic coelom

27. Normal fertilization of the human oocyte by a spermatozoon typically takes place within which anatomical region of the uterine tube?

a. Isthmus
b. Infundibulum

c. Fimbriae
d. Intramural part
e. Ampulla

28. When fetal capillaries and blood cells successfully differentiate within the internal mesodermal core of a chorionic villus, it becomes a:

a. Primary villus
b. Secondary villus
c. Tertiary villus
d. Stem villus
e. Anchoring villus

29. The pre-embryonic cluster of cells differentiates into an outer cell layer that forms the protective membranes and an inner cell mass known as the:

a. Trophoblast
b. Embryoblast
c. Blastocoel
d. Morula
e. Blastocyst

30. The second week of development is characterized by the systemic replication of structures into pairs, a clinical concept often taught as the:

a. Rule of Twos
b. Principle of Bilaterality
c. Dual Layer Phase
d. Gastrulation Axis
e. Somite Sequence

Answer Key

Q 1-8	Q 9-16	Q 17-24	Q 25-30

e	**b**	**b**	**d**
d	**b**	**a**	**b**
d	**a**	**e**	**e**
e	**c**	**a**	**c**
a	**b**	**d**	**b**
c	**c**	**a**	**a**
c	**e**	**d**	
e	**e**	**c**	